Ref. [54] — Escaping Higher-Order Saddle Points

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1. What goes wrong with Hessian-based saddle escape?

Perturbed gradient descent and negative-curvature methods ask whether the minimum Hessian eigenvalue is significantly negative. But for a degenerate saddle, even the Hessian can be positive semidefinite. The authors ask whether third derivatives provide a principled escape method, and whether the same idea can continue indefinitely.

2. Rebuild the problem using a simple mathematical example

Consider the two-variable function

$$f(x, y) = x^2 - y^4.$$

At $(0, 0)$, the gradient vanishes and the Hessian is $\mathrm{diag}(2, 0)$; no negative quadratic curvature is visible. Yet $f(0, y) = -y^4 < 0$ when $y \neq 0$. This *fourth-order* example is for intuition: a second-order test can miss descent even when the function is not locally minimized. For a smooth function and a displacement u, Taylor expansion shows the additional information available at third order:

$$f(x + u) = f(x) + \nabla f(x)^\top u + \tfrac{1}{2} u^\top \nabla^2 f(x) u + \tfrac{1}{6} \nabla^3 f(x)[u, u, u] + R_4.$$

The tensor $\nabla^3 f(x)[u, u, u]$ is a cubic directional derivative. It can detect escape directions unavailable to the gradient/Hessian pair. A simple *third-order* saddle example is $h(x, y) = x^2 + y^3$: at the origin, the Hessian is semidefinite, but the y-direction changes sign across the origin.

3. Actual algorithmic and hardness results

The paper constructs an efficient procedure to find an approximate *third-order local optimum* under its smoothness and oracle assumptions. It combines first-/second-order information with tests using higher-order derivatives to avoid certain degenerate stationary regions. Crucially, the authors show that extending this program to finding a *fourth-order local optimum* is NP-hard in the general class they study. This separates two tasks:

$$\underbrace{\text{second-order stationarity}}_{\text{no strong negative Hessian direction}} \not\Rightarrow \underbrace{\text{third-order local optimality}}_{\text{also rules out relevant cubic descent}}.$$

Paper result. There is a provable algorithm for reaching third-order local optimality, but no comparable unrestricted polynomial-time method for fourth-order local optimality unless standard complexity assumptions fail.

4. Why this matters for deep linear networks

For a toy deep linear network $\hat{y} = abc$ and target 1, take

$$L(a, b, c) = \tfrac{1}{2}(1 - abc)^2.$$

At $(0, 0, 0)$, both the gradient and Hessian vanish, but along (t, t, t):

$$L(t, t, t) = \tfrac{1}{2} - t^3 + \tfrac{1}{2} t^6.$$

The cubic directional term makes the loss smaller for small $t > 0$. This *teaching example* links Kawaguchi's degenerate deep-linear saddles to the paper's central idea; it is not presented as one of Anandkumar and Ge's experiments.

5. Assumptions, limits and relevance

These are general smooth, higher-order oracle results with regularity and accuracy parameters; they do not yield a cheap practical training routine for large Transformers, where third-order derivative calculations and activation kinks are challenging. A third-order escape guarantee also does not imply global optimality. The paper is central to **RQ1**, **RQ3** and **RQ4** because it exposes a limitation of the usual strict-saddle paradigm.

Scope. The NP-hardness claim concerns general fourth-order local optimality, not the impossibility of training all deep neural networks. Structured neural-network degeneracies may be more tractable.

6. Our notes

I like this paper as the logical follow-up to Kawaguchi. If a saddle has no negative Hessian eigenvalue, an optimizer looking only at curvature might think it is safe to stop. Third derivatives can reveal a descent direction, but trying to extend this to every higher order quickly hits a complexity barrier. It explains why our survey shouldn't equate SOSP with a true minimum. **Original source:** https://proceedings.mlr.press/v49/anandkumar16.html. **Verification:** official COLT abstract and original manuscript checked; toy examples labelled explicitly.